
TN 19.9.0.0: First/Last Surface Extraction

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1 Challenge: Interference Envelope of Transparent Samples

For a single scan of transparent and semi-transparent samples a pixel's interference envelope comprises one peak per reflective interface in the scan range (» Fig. 1.1).

A single surface is, hence, not defined unambiguously. However, the application may either require measuring the first surface consistently or the last surface consistently.

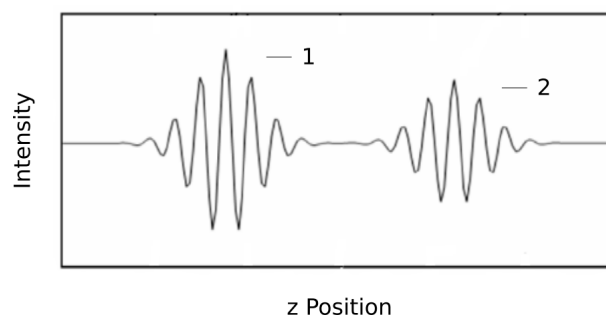


Fig. 1.1: Sample comprising a transparent layer: Typical intensity waveform of a pixel plotted in the z-direction with (1) first surface and (2) second surface

2 Solution: Amplitude Peak Detection and Thresholding

- Use `PeakDetectionMode` to distinguish first and last occurrences of amplitude maxima along `z` in scan direction. Relevant entries are: `FirstMax`, `LastMax`, `FilteredFirstMax`, `FilteredLastMax` (The latter two use amplitude peak filtering. For more details refer to » vertical-filtering-amplitude-peak).

Note: `PeakDetectionMode` configures the algorithm used for coarse surface detection. It has no influence when `Scan3dExtractionMethod` is `rawIQ` or `rawAmp`.

- If selecting `FilteredFirstMax` or `FilteredLastMax`, define a value for `PeakDetectionFilterWidth`.
- Define a suitable amplitude threshold value to enable the device to separate noise from the relevant signal and write it to `PeakDetectionThreshold`. Iteratively optimize the threshold value. Noise-induced surface outliers appear at random height if the value is too low. No surface is detected (pixel's `z` data set to end of scanned range) if the threshold value is too high.

Note: A good starting point for `PeakDetectionThreshold` can be found by first measuring in `rawAmp` mode and selecting an amplitude value between the noise floor and the envelope peak of interest. A good starting point for the `PeakDetectionFilterWidth` is the width of the signal amplitude peak. Filtering slightly reduces peak amplitude. Reduce the value of `PeakDetectionThreshold` accordingly.

3 Tips for Narrow Transparent Layers

When peak separation is a small multiple of peak width:

- Use minimum vertical spacing (`TargetVerticalSpacing` equal to wavelength if `BackgroundSuppression` = 'off', $1.5 \times \text{wavelength}$ otherwise)
- Use unfiltered surface detection if peak amplitude is consistently well above noise level (`FirstMax` or `LastMax`).
- When using filtered surface detection (`FilteredFirstMax`, `FilteredLastMax`), use `PeakDetectionFilterWidth` values lower than the peak separation (in unit of frames)
- Use `ExtSimpMaxHWin` values lower than the peak separation (in units of frames). `ExtSimpMaxHwin` defines the vertical size of the data window (around the detected peak, in unit of frames) used for (fine) surface extraction. If using a larger data window size, the peak is not properly isolated and adjacent peaks may alter the fine surface result.

4 Implementation

```
# Enable first surface detection across the a-scan.  
PeakDetectionMode = "FirstMax"  
  
# Configure amplitude threshold  
PeakDetectionThreshold = 84
```

5 Document History

Version	Date	Changes	Responsible
19.9.0.0	07-04-2025	Initial version	cl